Hydrogen held up by the zero-point field

Does the orbiting electron absorb from the zero-point field exactly what it radiates, and does that fix hydrogen's ground state? (Puthoff 1987) — register entry H24

kronspice simulation report · 2026-11-07 · CODATA 2018; 16 random-phase runs

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Summary

H. E. Puthoff (Phys. Rev. D 35, 3266, 1987) works in **stochastic electrodynamics** (SED), which treats empty space as filled with a real, random classical electromagnetic field carrying ½ħω of energy per mode, the zero-point field. A classical electron circling a proton radiates and should spiral in within about 16 ps. Puthoff shows that on the orbit with $m \cdot \omega \cdot r^2 = \hbar$ (m the electron mass, ω the orbit's angular frequency, r its radius) the electron absorbs from the zero-point field exactly what it radiates, so it stays. This example recomputes the paper's balance with CODATA values, checks whether the balance is restoring, simulates the paper's oscillator in a random zero-point field, and builds the same oscillator as a kronspice circuit driven by its resistor's zero-point noise.

- **The balance gives Bohr's numbers:** r = 5.291772×10^{-11} m (the Bohr radius a_0) and −13.6057 eV, with 4.669288×10^{-8} W absorbed and the same radiated. This is expected: the paper's eq. 20 is Bohr's condition.
- **The averaged balance is restoring:** absorbed/radiated is 1.414 at 0.5 a_0 and 0.707 at 2 a_0, pulling the orbit back to a_0 with a time constant of 9.34×10^{-11} s (≈ 614,349 orbits).
- **The oscillator in a simulated zero-point field holds ½ħω_0:** mean energy 0.5037 ± 0.0009 ħω_0 (expected 0.5036); absorbed 0.005076 and radiated 0.005077 agree to 0.03 %.
- **The kronspice tank in zero-point resistor noise** stores 0.4916 ± 0.0035 ħω_0 (expected 0.4935); its ledger closes (noise source 645.0 J = resistor 639.3 J + stored 4.59 J, mismatch 0.002).
- **Zero net energy exchange:** what the zero-point field puts in comes back out. Nothing here is an energy source.
- **Status:** reproduced at Bohr-theory level, as the paper claims; beyond it untested here.

The hypothesis and the standard prediction

H24 (hypothesis): hydrogen's ground state is a zero-point-fluctuation-determined state: in SED the orbiting electron absorbs from the random zero-point field exactly what it radiates when $m \cdot \omega \cdot r^2 = \hbar$.

Standard prediction: the quantum-mechanical ground state has zero orbital angular momentum and a spread of radii (most probable a_0, mean 1.5 a_0); a single circular orbit with angular momentum ħ is not it. Puthoff's result is at the level of Bohr theory, as he says; extending SED to the Schrödinger ground state was "yet to be achieved" in 1987 (his footnote 17).

Results

1. The paper's numbers with CODATA 2018 values

Quantity	Result	Compare
Orbit radius from $m \cdot \omega \cdot r^2 = \hbar$ and Coulomb (eq. 20)	5.291772×10^{-11} m	Bohr radius a_0 = 5.291772×10^{-11} m
Energy	−13.6057 eV	Bohr and Schrödinger, infinitely heavy proton: −13.6057 eV (measured ionization energy 13.598 eV includes the proton's recoil)
Power absorbed from the zero-point field (eq. 18)	4.669288×10^{-8} W	radiated (eq. 19): 4.669288×10^{-8} W
Orbit frequency	6.5797×10^{15} Hz	
Collapse time with no zero-point field	1.56×10^{-11} s	the classical problem Puthoff answers

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2. Does the balance hold the orbit?

Net power $P_{abs} - P_{rad}$ on Coulomb orbits of other radii, averaged as in the paper:

Radius	Absorbed ÷ radiated	Net power
0.5 a_0	1.414	$+3.095\times10^{-7}$ W (gains energy, grows)
0.7 a_0	1.195	$+3.797\times10^{-8}$ W
0.9 a_0	1.054	$+3.850\times10^{-9}$ W
1.0 a_0	**1.000**	**-2.847×10^{-17} W**
1.1 a_0	0.953	-1.484×10^{-9} W
1.5 a_0	0.816	-1.693×10^{-9} W
2.0 a_0	0.707	-8.548×10^{-10} W (loses energy, shrinks)

The ratio absorbed/radiated is $\hbar/(m\omega r^2)$, which on Coulomb orbits falls as $1/\sqrt{r}$ (0.707 at 2 a_0, 1.414 at 0.5 a_0). The averaged balance pulls the orbit back to a_0 with a time constant of 9.34×10^{-11} s (≈ 614,349 orbits).

3. The paper's oscillator in a simulated zero-point field

Puthoff builds the orbit from two one-dimensional oscillators (eqs. 4–17). Here that oscillator is simulated directly: a random field with the zero-point spectrum is made from random phases (eq. 2) and a Fourier transform, and the radiating oscillator is integrated in it (own Runge–Kutta), 16 independent runs.

	Simulated	Expected for this field	Formula
Mean energy	0.5037 ± 0.0009 $\hbar\omega_0$	0.5036	$\frac{1}{2}\hbar\omega_0 = 0.5000$
Power absorbed	0.005076	0.005076	eq. 16: $\hbar\Gamma\omega_0^3/2 = 0.005000$
Power radiated	0.005077		

In steady state the oscillator radiates what it absorbs (difference 0.03 %) and holds $\frac{1}{2}\hbar\omega_0$, the known SED result for a harmonic oscillator (Boyer 1975). Declared choices:

- **Units:** scaled, $m = \omega_0 = \hbar = 1$.
- **Damping:** $\Gamma\omega_0 = 0.01$ instead of the electron's 2.6×10^{-7} at the hydrogen orbit frequency, so a run spans 524 relaxation times. Every result is compared with the formula for that damping.
- **Field cut:** the field stops at $2\omega_0$. The full zero-point spectrum makes a classical particle's kinetic energy diverge.
- **Radiation reaction:** used in its reduced (Landau–Lifshitz) form, which is equal at resonance.

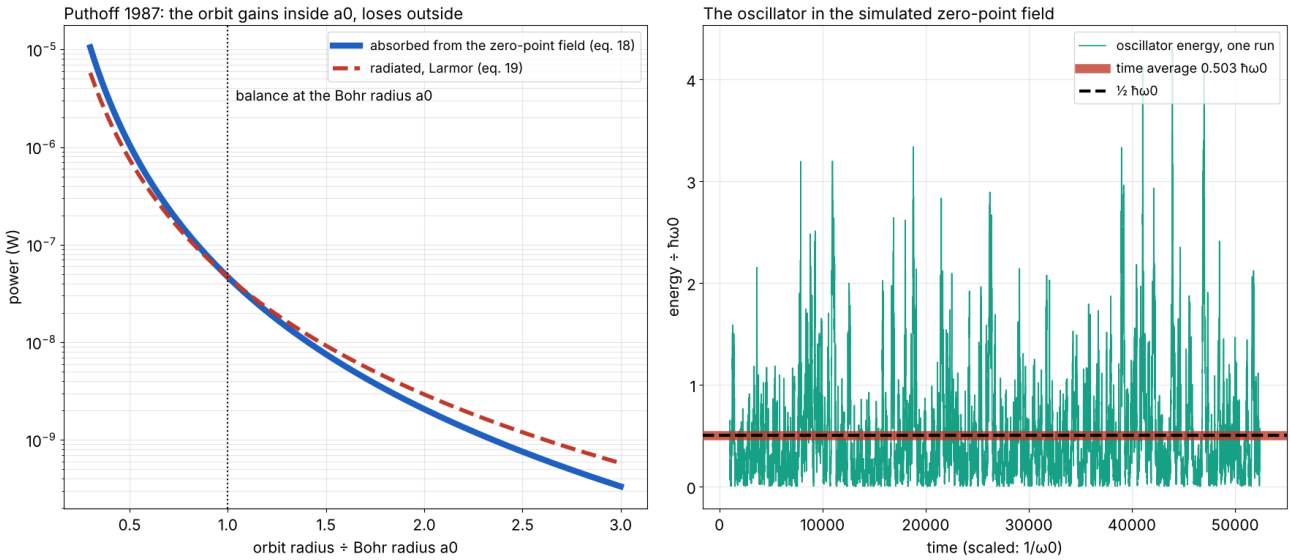


Figure 1. Left: power absorbed from the zero-point field (eq. 18, blue) and radiated (Larmor, eq. 19, red dashed) on Coulomb orbits from 0.3 to 3 a_0; they cross at the Bohr radius, with absorption winning inside and radiation outside. Right: the oscillator's energy in one simulated zero-point-field run (scaled time $1/\omega_0$); its time average, 0.503 $\hbar\omega_0$, lies on $\frac{1}{2}\hbar\omega_0$.

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4. The same oscillator as a kronspice circuit

A series tank of L = 1 H, C = 1 F and R = 0.01 Ω is driven by the resistor's own zero-point noise. By Callen and Welton's fluctuation–dissipation theorem (1951, cited in Puthoff's footnote 13), a resistor at absolute zero has a one-sided voltage noise $S_V = \hbar\omega R/\pi$ per unit angular frequency. ħ is scaled to 1 J·s; the circuit is linear, so energies scale with ħ.

	kronspice	Expected
Energy stored in the tank	0.4916 ± 0.0035 $\hbar\omega_0$	0.4935
Ledger	noise source 645.0 J = resistor 639.3 J + stored 4.59 J	mismatch 0.002

- The expected value allows for kronspice joining the noise waveform's points by straight lines.
- A kronspice lesson from this run: after every corner of a stepwise (PWL) waveform kronspice takes a short backward-Euler step (standard SPICE practice). With 65,536 corners that step damped the tank about as much as its resistor did, and the energy came out 7 % low. A maximum step of 0.02 s makes the extra damping negligible.
- The noise source puts in what the resistor takes out: net, the zero-point field supplies nothing, as in Puthoff's balance.

Where this stands

- **Bohr-theory level.** Eq. 20 is Bohr's condition, so energy and size agree with Bohr and Schrödinger. The test of SED is everything beyond that.
- **Later full SED simulations:** Cole & Zou (Phys. Lett. A, 2003) report the Schrödinger ground-state probability density; Nieuwenhuizen & Liska (Phys. Scr. T165, 014006, 2015), with longer runs, find that the atom eventually ionizes itself.
- **Energy.** The zero-point exchange is zero net: absorbed equals radiated.

Ledger and validation

- Paper's balance: absorbed and radiated power equal at a_0 (net -2.847×10^{-17} W).
- Oscillator: absorbed and radiated agree to 0.03 %; mean energy matches the expectation for the cut field.
- kronspice tank: ledger noise source = resistor + stored to a mismatch of 0.002.
- Tests (*main_test.go*): Bohr numbers, stability ratio, oscillator energy and balance, tank energy and ledger, FFT.

Limits of the model

- A single circular orbit, averaged as in the paper; no orbital spread and no full three-dimensional SED atom.
- The oscillator uses a damping 0.01 instead of 2.6×10^{-7} and a field cut at $2\omega_0$ (declared).

Status

H24 is **reproduced at Bohr-theory level, as the paper claims; beyond it untested here.** The quantum ground state has L = 0 and a spread of radii; later full SED simulations report the Schrödinger density (Cole & Zou) and, with longer runs, self-ionization (Nieuwenhuizen & Liska). The zero-point exchange is zero net, not an energy source.

Reproducing the results

```
go run ./examples/zpf_hydrogen -viz examples/zpf_hydrogen/viz   # about 15 seconds
python3 examples/zpf_hydrogen/viz/render.py
go test ./examples/zpf_hydrogen
```

Sources: H. E. Puthoff, Phys. Rev. D 35, 3266 (1987); T. H. Boyer, Phys. Rev. D 11, 790 (1975); H. B. Callen, T. A. Welton, Phys. Rev. 83, 34 (1951); D. C. Cole, Y. Zou, Phys. Lett. A (2003), doi:10.1016/j.physleta.2003.08.022; T. M. Nieuwenhuizen, M. T. P. Liska, Phys. Scr. T165, 014006 (2015); CODATA 2018.

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